Practice #Final *<100 pts>*

1) For each of the following statements, is it TRUE or FALSE? *<10>*

a. The <u>non-zero</u> eigenvalues of $\boldsymbol{AA}^T$ are the same as those of $\boldsymbol{A}^T\boldsymbol{A}$.

b. If an SVD of a <u>symmetric</u> matrix $\boldsymbol{A}$ is $\boldsymbol{U\Sigma V}^T$, then $\boldsymbol{U} = \boldsymbol{V}$.

c. $\boldsymbol{AA}^+\vec{x} = \vec{x}$ if and only if $\vec{x} \in R(\boldsymbol{A})$.

d. There exists a linear map $T: \mathbb{R}^n \to \mathbb{R}^n$ that is one-to-one but not onto.

e. The determinant of $\boldsymbol{A}^+\boldsymbol{A}$ is either 0 or 1.

f. For a <u>singular</u> matrix $\boldsymbol{A}$, $N(\boldsymbol{A})$ is one of the eigenspaces of $\boldsymbol{A}$.

g. If a matrix $\boldsymbol{A}$ represents a map $T: \mathbb{R}^m \to \mathbb{R}^n$, then $\dim\big(N(\boldsymbol{A}^T)\big) \geq m - n$.

h. One eigenvalue of $\boldsymbol{A}_{n\times n}$ is k if the sum in every row is equal to k.

i. If the singular values of $\boldsymbol{A}^{-1}$ are 0.5 and 2, then $|0.5\vec{x}| \leq |\boldsymbol{A}\vec{x}| \leq |2\vec{x}|$.

j. The determinant $\boldsymbol{A}^+\boldsymbol{A}$ of $\boldsymbol{AA}^+$ and are <u>always</u> equal.

2) Let $\boldsymbol{A} = \begin{bmatrix} 9 & -24 & 0 \\ 12 & 6 & -15 \\ 0 & -10 & 5 \end{bmatrix}$ and $\boldsymbol{A} = \boldsymbol{UD}$ where $\boldsymbol{D} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$, find $\boldsymbol{U}$. *<2>*

3) Let $\boldsymbol{A} = \begin{bmatrix} 1 & x \\ 0 & 1+x \end{bmatrix}$ where $x \neq 0$, what is the angle between the eigenvectors of $\boldsymbol{A}$? <2>

4) Let $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & x \\ -1 & 1 & y \\ -1 & 2 & xy+2 \end{bmatrix}$, if $\boldsymbol{A}$ is singular, what is the value of x and y? <2>

5) Let $\boldsymbol{A} = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$, <4>

a. An orthonormal basis for $R(\boldsymbol{A})^{\perp}$ = _____. <1>

b. All singular values of $\boldsymbol{A}$ = _____. <1>

c. $(\boldsymbol{A}\boldsymbol{A}^{+})^{101}$ = _____. <1>

d. $\boldsymbol{A}^{+}\boldsymbol{A}\begin{bmatrix} 3 \\ 10 \\ -3 \end{bmatrix}$ = _____. <1>

6) Let $\boldsymbol{A}$ be a matrix which is diagonalized as $\begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}\begin{bmatrix} -15 & 0 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}^{T}$, <7>

a. SVD of $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{T}$; $\boldsymbol{U}$ = _____, $\boldsymbol{\Sigma}$ = _____, and $\boldsymbol{V}$ = _____. <3>

b. SVD of $\boldsymbol{A}^{-1} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{T}$; $\boldsymbol{U}$ = _____, $\boldsymbol{\Sigma}$ = _____, and $\boldsymbol{V}$ = _____. <3>

c. If $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{T}$ where $\boldsymbol{D} = \begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix}$, then $\boldsymbol{P}$ = _____. <1>

7) Let $\boldsymbol{A} = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & -3 \\ 1 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, find an orthonormal basis for $C(\boldsymbol{A})$. <5>

8) Let $\boldsymbol{A}$ be an m-by-n matrix, <4>

a. Calculate $\dim\big(C(\boldsymbol{A})\big) + \dim(R(\boldsymbol{A}^T)^{\perp}) + \dim(C(\boldsymbol{A}^T)^{\perp}) + \dim(N(\boldsymbol{A})^{\perp})$. <2>

b. Calculate $\dim(R(\boldsymbol{A})^{\perp}) - \dim\big(N(\boldsymbol{A}^T)\big)$. <2>

9) Let $\boldsymbol{A}$ be a matrix whose SVD is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}$, <6>

a. The rank of $\boldsymbol{A}$ = _____, the nullity of $\boldsymbol{A}$ = _____,

a basis for $R(\boldsymbol{A})$ = _____, a basis for $N(\boldsymbol{A})$ = _____,

a basis for $C(\boldsymbol{A})$ = _____, a basis for $N(\boldsymbol{A}^T)$ = _____. <3>

b. SVD of $\boldsymbol{A}^{+} = \boldsymbol{U\Sigma V}^T$; $\boldsymbol{U}$ = _____, $\boldsymbol{\Sigma}$ = _____, and $\boldsymbol{V}$ = _____. <3>

10) Let $\boldsymbol{A} = \begin{bmatrix} 0 & 2 & -2 & 2 & -2 & -2 \\ 0 & -1 & 3 & -1 & 2 & 1 \\ 0 & 2 & 2 & 2 & -1 & 0 \end{bmatrix}$ and $RRE(\boldsymbol{A}) = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 \end{bmatrix}$, <5>

a. Which columns of $\boldsymbol{A}$ form a basis for $C(\boldsymbol{A})$? <1>

b. Find a smallest set that spans the orthogonal complement of $R(\boldsymbol{A})$. <2>

c. Find $[\vec{a}_4 \quad \vec{a}_3 \quad \vec{a}_5]^{-1}[\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_6]$, where $\vec{a}_i$ is the ith column of $\boldsymbol{A}$. <2>

11) Find one eigenvalue and its associated eigenvector of $\begin{bmatrix} 32 & 18 & 23 & 11 & 16 \\ 10 & 20 & 40 & 20 & 10 \\ 24 & 21 & 10 & 21 & 24 \\ 0 & 10 & 20 & 30 & 40 \\ 34 & 31 & 7 & 18 & 10 \end{bmatrix}$. <2>

12) If $\begin{bmatrix} a & b & c \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{bmatrix}\begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix} + 9\boldsymbol{I} = \boldsymbol{0}$, what is the length of $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$? <2>

13) Let $S = \{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}\}$, find the largest independent subset of S. <5>

14) Find $\text{span}(\{\begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\\2\end{bmatrix}\})$ as a perp. <2>

15) Find a diagonalization of the following matrices: <7>

a. $\boldsymbol{A} = \begin{bmatrix}2 & 1\\3 & 4\end{bmatrix}$ <3>

b. $\boldsymbol{A} = \begin{bmatrix}2 & 0 & 0\\3 & -1 & 0\\1 & -2 & -3\end{bmatrix}$ <4>

16) Find an orthogonal diagonalization of $\boldsymbol{A} = \begin{bmatrix}0 & -1 & 1\\-1 & 0 & -1\\1 & -1 & 0\end{bmatrix}$. <5>

17) Let $\boldsymbol{A} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$, <8>

a. Find an SVD of $\boldsymbol{A}$. <5>

b. Find $\boldsymbol{A}^{+}$. <3>

18) Let $\boldsymbol{A}$ be a square matrix whose diagonal entries $a_{i,i} = d$ and other entries $a_{i,j} = x$ where $x, d \in \mathbb{R}$ and $x \neq 0$, that is $\boldsymbol{A}_{n\times n} = \begin{bmatrix} d & x & x & \dots & x \\ x & d & x & \dots & x \\ x & x & d & \dots & x \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x & x & x & \dots & d \end{bmatrix}$, find the spectrum of $\boldsymbol{A}$ and all eigenspaces of $\boldsymbol{A}$. <5>

19) Let $\boldsymbol{A} = \begin{bmatrix} -5 & 6 \\ -3 & 4 \end{bmatrix}$, calculate $\det(\boldsymbol{A}^{12} - 5\boldsymbol{A}^{10} - 3\boldsymbol{A}^{9} + 12\boldsymbol{A}^{5} - 6\boldsymbol{A})$. <5>

20) Given square matrices $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{S} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$ such that $\boldsymbol{SA} - \boldsymbol{BS} = \boldsymbol{0}$, one of the eigenvalues of $\boldsymbol{B}$ is 3, a basis for $N(\boldsymbol{A}^T)$ is $\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right\}$: <12>

a. Rank of $\boldsymbol{B}$ = _____, nullity of $\boldsymbol{B}$ = _____, and $\det(\boldsymbol{B})$ = _____. <3>

b. Find $C(\boldsymbol{B})$ as a span. <4>

c. If $\boldsymbol{A}$ is symmetric and $\boldsymbol{A}\begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix}$, find $\boldsymbol{A}$. <5>

Answers:

1) TRUE: a. e. f. h. i.

FALSE: b. c. d. g. j.

2) $\boldsymbol{U} = \begin{bmatrix} 3 & -12 & 0 \\ 4 & 3 & -3 \\ 0 & -5 & 1 \end{bmatrix}$

3) $\frac{\pi}{4}$ or $45°$

4) $x = 2, y = 1$

5) a. $\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right\}$ b. $\sqrt{2}$ c. $[1]$ d. $\begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$

6) a. $\boldsymbol{U} = \begin{bmatrix} -\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} 15 & 0 \\ 0 & 5 \end{bmatrix}, \boldsymbol{V} = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$

b. $\boldsymbol{U} = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{15} \end{bmatrix}, \boldsymbol{V} = \begin{bmatrix} -\frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix}$

c. $\boldsymbol{P} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

7) $\left\{\frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{3}\begin{bmatrix} 2 \\ 0 \\ 2 \\ -1 \end{bmatrix}\right\}$

8) a. $m + n$ b. $n - m$

9) a. 2, 1, $\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}$, $\left\{\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right\}$, $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$, $\emptyset$

b. $\boldsymbol{U} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}, \boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & 0 \end{bmatrix}, \boldsymbol{V} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

10) a. $2^{nd}, 3^{rd}, 5^{th}$

b. $\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}\right\}$

c. $\begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{bmatrix}$

11) $\lambda = 100, \vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

12) 3

13) $\left\{\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 6 \end{bmatrix}\right\}$

14) $\text{perp}\left(\left\{\begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix}\right\}\right)$

15) a. $\begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & 3 \end{bmatrix}^{-1}$

b. $\begin{bmatrix} -5 & 0 & 0 \\ -5 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix}\begin{bmatrix} -5 & 0 & 0 \\ -5 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}^{-1}$

16) $\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}^T$

17) a. $\boldsymbol{A} = \boldsymbol{U\Sigma V}^T$ where $\boldsymbol{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$,

$\boldsymbol{\Sigma} = \begin{bmatrix} \sqrt{8} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $\boldsymbol{V} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

b. $\boldsymbol{A}^+ = \frac{1}{8}\begin{bmatrix} 1 & 2 & -1 \\ -1 & 2 & 1 \\ 1 & 2 & -1 \\ -1 & 2 & 1 \end{bmatrix}$

18) $\text{spec}(\boldsymbol{A}) = \{d + (n-1)x, d - x\}$,

$E_{d+(n-1)x} = \text{span}\left(\left\{\begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}\right\}\right)$,

$E_{d-x} = \text{span}\left(\left\{\begin{bmatrix} -1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} -1 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}\right\}\right)$

19) -140

20) a. 2, 1, 0

b. $\text{span}\left(\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}\right)$

c. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

Review: Any matrix $\boldsymbol{A}_{m\times n}$ always represents some <u>linear</u> map $T:\mathbb{R}^n \to \mathbb{R}^m$ where $\boldsymbol{A}\vec{x} = T(\vec{x}) = \vec{y}$

Let $V = \text{span}(S)$, V is a subspace of $\mathbb{R}^n$ and $\dim(V) \le |S|$,

- $\dim(V) = |S|$ ***IFF*** S is linearly independent ***IFF*** S is a basis for V.
- $S = V$ ***IFF*** S is already a subspace.

Let S be a basis for $V \subseteq \mathbb{R}^n$, V is a subspace of $\mathbb{R}^n$ and $\dim(V) = |S| \le n$,

- $\dim(V) = n$ ***IFF*** $V = \mathbb{R}^n$ ***IFF*** S is a basis of $\mathbb{R}^n$; $\dim(V) = 0$ ***IFF*** $V = \{\vec{0}\}$ ***IFF*** $S = \emptyset$.
- $V^\perp$ is also a subspace of $\mathbb{R}^n$; $\dim(V) + \dim(V^\perp) = n$ and $V \cap V^\perp = \{\vec{0}\}$,
  - $S \cup T$ is a basis for $\mathbb{R}^n$ ***IFF*** T is a basis for $V^\perp$.

Let $\boldsymbol{A}_{m\times n}$ represent a mapping $T:\mathbb{R}^n \to \mathbb{R}^m$, four fundamental subspaces $R(\boldsymbol{A}), N(\boldsymbol{A}), C(\boldsymbol{A}), N(\boldsymbol{A}^T)$,

- $R(\boldsymbol{A}), N(\boldsymbol{A}) \subseteq \mathbb{R}^n$, $C(\boldsymbol{A}), N(\boldsymbol{A}^T) \subseteq \mathbb{R}^m$, $R(\boldsymbol{A}) = C(\boldsymbol{A}^T)$.
- $\#\text{pivots}(\text{RRE}(\boldsymbol{A})) = \#\text{pivots}(\text{RRE}(\boldsymbol{A}^T)) = \text{rank}(\boldsymbol{A}) = \text{rank}(\boldsymbol{A}^T) = \dim(R(\boldsymbol{A})) = \dim(C(\boldsymbol{A}))$.
- $\#\text{frees}(\text{RRE}(\boldsymbol{A})) = \text{nullity}(\boldsymbol{A}) = \dim(N(\boldsymbol{A}))$; $\#\text{frees}(\text{RRE}(\boldsymbol{A}^T)) = \text{nullity}(\boldsymbol{A}^T) = \dim(N(\boldsymbol{A}^T))$.
- $R(\boldsymbol{A}) = N(\boldsymbol{A})^\perp$, $\dim(R(\boldsymbol{A})) + \dim(N(\boldsymbol{A})) = n = \#\text{columns}(\boldsymbol{A})$.
- Combining a basis for $R(\boldsymbol{A})$ with a basis of $N(\boldsymbol{A})$ yields a basis for $\mathbb{R}^n$.
- $C(\boldsymbol{A}) = N(\boldsymbol{A}^T)^\perp$, $\dim(C(\boldsymbol{A})) + \dim(N(\boldsymbol{A}^T)) = m = \#\text{rows}(\boldsymbol{A})$.
- Combining a basis for $C(\boldsymbol{A})$ with a basis of $N(\boldsymbol{A}^T)$ yields a basis for $\mathbb{R}^m$.
- $\dim(N(\boldsymbol{A})) > 0$ ***IFF*** T is not one-to-one, $\dim(N(\boldsymbol{A}^T)) > 0$ ***IFF*** T is not onto.

<u>Projection of $\vec{v} \in \mathbb{R}^n$ onto a subspace $S \subseteq \mathbb{R}^n$</u>: (If $\dim(S) = m$)

- Let $\boldsymbol{A}_{n\times m}$ be a matrix whose columns is a <u>basis</u> for S, $\text{proj}_S(\vec{v}) = \boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\vec{v} \in S$.
- $\text{proj}_{S^\perp}(\vec{v}) = \vec{v} - \text{proj}_S(\vec{v})$ and $\vec{v} - \text{proj}_S(\vec{v}) \perp \text{proj}_S(\vec{v})$.
- $\vec{v} = \vec{x} + \vec{y}$ and $\vec{x} \perp \vec{y}$ ***IFF*** $\exists S \subseteq \mathbb{R}^n$: $\vec{x} = \text{proj}_S(\vec{v})$ and $\vec{y} = \text{proj}_{S^\perp}(\vec{v})$.
- If $\dim(S) = 1$ ($S = span(\{\vec{a}\})$ is a line), then $\boldsymbol{A}_{n\times 1}$ is just a column vector ($\boldsymbol{A}_{n\times 1} = \vec{a}$),
  - $\text{proj}_S(\vec{v}) = \vec{a}(\vec{a}^T\vec{a})^{-1}\vec{a}^T\vec{v} = \vec{a}(\vec{a}\cdot\vec{a})^{-1}(\vec{a}\cdot\vec{v}) = \frac{(\vec{a}\cdot\vec{v})}{(\vec{a}\cdot\vec{a})}\vec{a}$.

<u>Finding Orthonormal Basis</u> (Gram-Schmidt Process):

- Given a basis $B = \{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n\}$ for S, find an orthonormal basis B' for S.
  - Let $\vec{v}_1 = \vec{b}_1$ and for $1 < k \le n$; $\vec{v}_k = \vec{b}_k - \sum_{i=1}^{k-1}\text{proj}_{\vec{v}_i}(\vec{b}_k) = \vec{b}_k - \sum_{i=1}^{k-1}\frac{(\vec{b}_k\cdot\vec{v}_i)}{(\vec{v}_i\cdot\vec{v}_i)}\vec{v}_i$.
  - An orthonormal basis $B' = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ where $\vec{u}_k = \frac{\vec{v}_k}{|\vec{v}_k|}$ for $1 \le k \le n$.

<u>Eigenvalue</u> (λ) of $\boldsymbol{A}_{n\times n}$: <u>Eigenvector</u> $\vec{v}_\lambda$: $\boldsymbol{A}\vec{v}_\lambda = \lambda\vec{v}_\lambda$, <u>Eigenspace</u> $E_\lambda = \{all\ \vec{v}_\lambda\} = N(\boldsymbol{A} - \lambda\boldsymbol{I})$,

- Characteristic polynomial: $\det(\boldsymbol{A} - \lambda\boldsymbol{I})$, Characteristic equation: $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$.
- $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = (\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2}\dots(\lambda - \lambda_k)^{m_k} = 0$; $\sum_{i=1}^k m_i = n$ (polynomial degree n).
- $\text{spec}(\boldsymbol{A}) = \{\lambda_1, \lambda_2, \dots, \lambda_k\}$, $\prod_{i=1}^k(\lambda_i)^{m_i} = \det(\boldsymbol{A})$, $\sum_{i=1}^k m_i\lambda_i = \text{trace}(\boldsymbol{A})$.
- Algebraic multiplicity: m_i, Geometric multiplicity: $\dim(E_{\lambda_i})$; $1 \le \dim(E_{\lambda_i}) \le m_i$.

$\boldsymbol{A}$ and $\boldsymbol{B}$ are <u>*similar*</u> ***IFF*** $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{B}\boldsymbol{P}^{-1}$; [<u>same</u>: det, spectrum, (both)multiplicity] but [<u>not same</u>: E_λ].

$\boldsymbol{A}$ is <u>*diagonalizable*</u> ***IFF*** $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$ for some diagonal matrix $\boldsymbol{D}$ ***IFF*** $\dim(E_{\lambda_i}) = m_i$,

- $\text{diag}(\boldsymbol{D})$: λ_i (repeat with multiplicity), $\boldsymbol{P} = [\vec{v}_{\lambda_i}]$ ($\vec{v}_{\lambda_i}$: vector basis of E_{λ_i}).
- Orthogonally diagonalizable ($\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T$) ***IFF*** $\boldsymbol{P}$ is <u>orthonormal</u> ***IFF*** $\boldsymbol{A}$ is symmetric.

<u>SVD</u> of $\boldsymbol{A}_{m\times n} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$: $\boldsymbol{U}_{m\times m}, \boldsymbol{V}_{n\times n}$ are orthonormal, $\boldsymbol{\Sigma}_{m\times n}$ is quasi-diagonal,

- $\text{diag}(\boldsymbol{\Sigma})$: σ_i ($\sigma_1 \ge \sigma_2 \ge \dots \ge 0$), σ_i are singular values of $\boldsymbol{A}$, $\text{rank}(\boldsymbol{A}) = \#$non-zero σ_i of $\boldsymbol{A}$.
- $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$ or $\boldsymbol{A}\boldsymbol{A}^T = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\boldsymbol{U}^T = \boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$, $\sigma_i = \sqrt{\lambda_i}$, $\sigma_i\vec{u}_i = \boldsymbol{A}\vec{v}_i$, $\sigma_i\vec{v}_i = \boldsymbol{A}^T\vec{u}_i$.
- If $\text{rank}(\boldsymbol{A}) = r$, basis of $C(\boldsymbol{A})$:$\{\vec{u}_1, \dots, \vec{u}_r\}$, $N(\boldsymbol{A}^T)$:$\{\vec{u}_{r+1}, \dots, \vec{u}_m\}$, $R(\boldsymbol{A})$:$\{\vec{v}_1, \dots, \vec{v}_r\}$, $N(\boldsymbol{A})$:$\{\vec{v}_{r+1}, \dots, \vec{v}_n\}$.

<u>Pseudo-inverse of $\boldsymbol{A}$</u>: ($\boldsymbol{A}^+$ where $\text{proj}_{R(\boldsymbol{A})} = \boldsymbol{A}^+\boldsymbol{A}$, $\text{proj}_{C(\boldsymbol{A})} = \boldsymbol{A}\boldsymbol{A}^+$),

- If an SVD of $\boldsymbol{A}_{m\times n} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, then $\boldsymbol{A}^+_{n\times m} = \boldsymbol{V}\boldsymbol{\Sigma}^+\boldsymbol{U}^T$, $\text{diag}(\boldsymbol{\Sigma}^+_{n\times m})$: σ_i^{-1} (if $\sigma_i > 0$) or 0 otherwise.